Project Brief

Phase I: Data Ingestion & Infrastructure

Context: Why This Matters

Sciencia AI is supporting a client in developing a next-generation, AI-powered analytics platform to understand and respond to user sentiment at scale. The goal is not simply to classify positive or negative feedback, but to build a deeper, structured understanding of **how users think, feel, and behave**—insights that directly inform product strategy, service design, and experience optimization.

To enable such intelligence, a performant model alone is not sufficient. What's needed is a **reliable, extensible, and self-improving data pipeline** that can continuously:

collect data → clean and structure it → label it → train models → evaluate and iterate.

This project focuses on building the **foundational layer** of that pipeline: the infrastructure responsible for ingesting user-generated textual data from the open web, transforming it into usable structure, and storing it in a way that downstream teams can leverage for labeling, training, and experimentation.

By laying down this core infrastructure, we unlock the ability to evolve from static, one-off analysis toward a **dynamic, production-grade learning system** that continuously improves as more data becomes available.

Role and Scope: Building the Entry Point of the Pipeline

The task at hand is to design and implement a **modular, maintainable system** for automated data acquisition and structured storage. This is the first step in transforming the open web into a structured knowledge asset for machine learning.

Rather than a quick scraping script or disposable tool, this system is expected to be:

- **Modular**: logically separated into configurable components (e.g., extraction, transformation, storage)
- **Reusable**: designed to support multiple runs, data sources, and configurations
- **Sustainable**: built with maintainability and extensibility in mind, supporting future team members and scale

This infrastructure will serve as the **entry point to an evolving machine learning workflow**, and its robustness will directly affect the success of downstream components such as labeling interfaces, training pipelines, model evaluation loops, and future automation.

It is expected to deliver three core capabilities:

- **Data acquisition**: Automating the extraction of relevant review data from public web sources
- **Data structuring**: Converting raw, semi-structured content into a clean format suitable for storage, search, and modeling
- **Data storage**: Organizing the information within a scalable, queryable relational database that can be integrated into future systems

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Key Components to Be Developed

To achieve this, the following components are in scope for this phase:

1. Identifying a suitable data source

The system should be built on top of a well-chosen, high-quality data stream. Key criteria for selection include:

- Sufficient **volume** of user-generated content (e.g., product reviews, community discussions)
- **Public accessibility** without login or gated APIs
- **Stable structure** (HTML layout or API format) suitable for automated parsing
- Clear **relevance to sentiment understanding**, such as reviews with ratings, timestamps, or user metadata

This decision will affect all downstream work, so it must balance feasibility, richness, and technical constraints.

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2. Developing an automated ingestion process

This module is responsible for programmatically retrieving content from the selected source, ideally through a script or framework that is:

- Modular and **configurable**, so that new sources or endpoints can be plugged in easily
- **Resilient** to pagination, content layout changes, and basic anti-scraping mechanisms

- Respectful of rate limits and terms of service

In future iterations, this ingestion layer could evolve into a scheduling or monitoring system that supports periodic updates and multi-source integration.

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3. Designing and implementing the database layer

Collected data needs to be persistently stored in a way that supports querying, analysis, and downstream consumption. This includes:

- Designing a **relational schema** that reflects the logical structure of the data (e.g., products, reviews, users)
- Balancing **normalization and performance**, so the data is organized and maintainable without sacrificing query efficiency
- Ensuring **compatibility with SQL-based analytics** and potential integration with model training pipelines

The choice of technology (e.g., SQLite for local prototyping vs PostgreSQL for deployment) will depend on the scale and accessibility requirements.

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4. Automating data cleaning and loading

After data is retrieved, it must be validated, cleaned, and stored in the database in a consistent and repeatable way. This includes:

- Handling **missing values, malformed fields, and non-standard encodings**
- Converting and normalizing fields such as timestamps, text, and categorical ratings
- Structuring the cleaned output and loading it into the designed schema
- Providing **basic verification queries or utilities** to inspect data health and completeness

While these tasks may seem operational, they are central to ensuring the **semantic integrity** of the dataset that will ultimately inform and shape the AI model's behavior.

This foundational system is not only critical to launching the broader pipeline—it also defines the **quality ceiling** of what the AI system can achieve. By investing in robust, reusable, and clean infrastructure, we enable:

- **Faster iteration cycles** for modeling and experimentation
- **Better interpretability** and traceability of AI decisions
- **Scalability** as additional data sources and feedback loops are introduced
- **Collaboration** between engineering, data science, and business stakeholders